

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile

devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for

redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the

access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

1. 20-Person Startup — Star Topology

Topology: Star topology

A star topology is ideal for a 20-person startup because every device connects to a central switch. A managed Layer 2/Layer 3 switch with at least 24 ports can connect workstations, printers, VoIP phones, and other devices while providing room for future expansion. Each device can be connected using Cat6 UTP cables, supporting Gigabit Ethernet up to 100 meters for typical office deployments.

The central switch connects to a router/firewall that provides Internet connectivity, DHCP, NAT, and security functions. A wireless access point can also be connected to the switch to provide Wi-Fi access.

Advantages:

- Easy to install and manage
- Simple troubleshooting
- Easy to expand
- Failure of one cable affects only one device
- Centralized network management

Justification:

If one workstation or its cable fails, the remaining devices continue to operate normally. Therefore, star topology provides good reliability, manageability, and scalability for a small startup.

2. Bank Requiring Near-Zero Downtime — Mesh Topology

Topology: Partial/Full Mesh topology

For a bank requiring very high availability, a mesh-based architecture is appropriate because it provides multiple communication paths. In a full mesh, every major network device has a direct connection to other devices. In practice, a **partial mesh** is often more practical and cost-effective.

Core routers, firewalls, and switches can have redundant connections, while access and distribution switches can use dual uplinks to separate core devices.

Fiber-optic links can be used for high-speed backbone connections.

Routing protocols such as **OSPF or BGP** can dynamically select alternative paths when a link fails.

Advantages:

- High fault tolerance
- Multiple backup paths
- Automatic rerouting
- High availability
- Suitable for critical financial applications

Justification:

If one network link fails, traffic can be redirected through another available path. Therefore, mesh topology is suitable for banking environments where network availability and reliability are critical.

3. Small Classroom Lab — Bus Topology

Topology: Bus topology

For a small classroom with a limited budget, a traditional bus topology can reduce infrastructure requirements because all computers share a common communication medium. In a classic Ethernet bus implementation, coaxial cable is used as the backbone, with appropriate connectors and terminators at both ends. However, traditional coaxial bus Ethernet is now largely obsolete. A modern classroom would normally use an inexpensive Ethernet switch and Cat5e/Cat6 UTP cables.

Advantages:

- Low initial hardware requirement in a traditional bus design
- Simple for a small network
- Suitable for light traffic

Limitations:

- Backbone failure can affect the entire network
- Traditional bus networks have collision issues
- Difficult to troubleshoot as the network grows

- Limited scalability

Justification:

For a small, supervised classroom with light network usage, a traditional bus topology can be considered when cost is the primary concern. However, a switched star topology would be the better modern implementation.

4. Large University Campus — Hybrid Topology

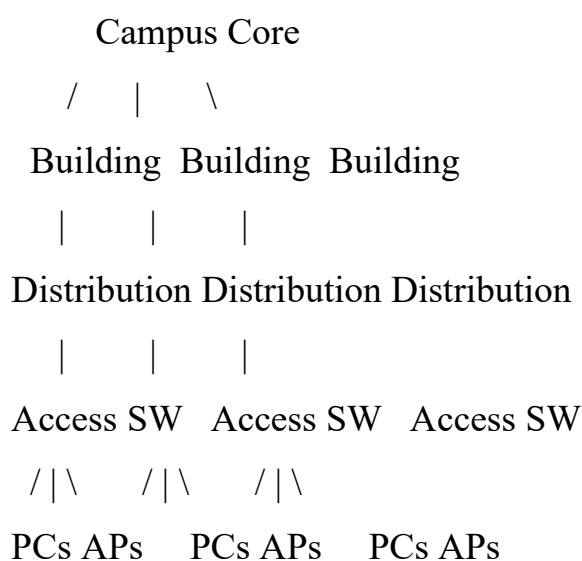
Topology: Hybrid topology

A large university campus requires a scalable architecture because it contains multiple buildings, laboratories, libraries, hostels, lecture halls, and administrative offices. Therefore, a **hybrid topology** combining different designs is appropriate.

At the campus backbone, high-speed fiber links can connect core routers and multilayer switches using a redundant ring or partial-mesh arrangement. Each building can have dual fiber uplinks to the campus core.

Inside each building, a hierarchical star topology can be used. Distribution switches connect to floor-level access switches, which then connect computers, printers, and wireless access points.

Architecture:



Advantages:

- Highly scalable
- Good fault tolerance
- Easy management

- Supports multiple buildings
- Easy addition of new users and buildings
- High-speed fiber backbone

Justification:

The hybrid design combines backbone redundancy with the simplicity of star topology inside buildings, making it suitable for a large university network.

5. Company with Multiple Departments — VLAN-Based Hierarchical Topology

Topology: Hierarchical star topology with VLANs

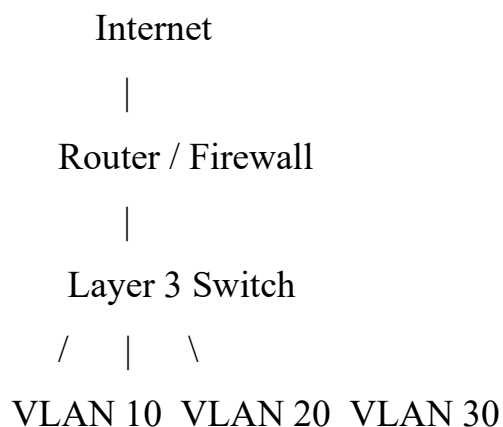
A company with departments such as HR, Finance, Sales, and IT can use a hierarchical network architecture. Each department connects its devices to an access-layer switch. VLANs logically separate the departments even when they use the same physical infrastructure.

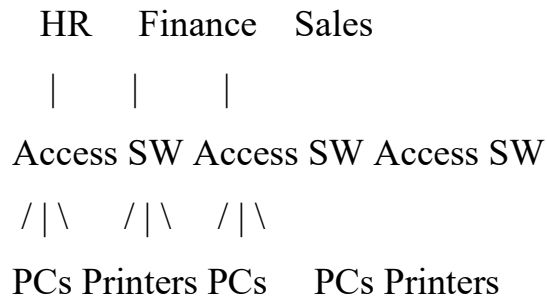
For example:

Department VLAN

HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40

A Layer 3 switch can perform **inter-VLAN routing**, allowing controlled communication between departments. A router/firewall connects the internal network to the Internet and provides security policies.



**Advantages:**

- Departmental network separation
- Better security
- Reduced broadcast traffic
- Centralized management
- Easy expansion
- Efficient IP addressing and subnetting
- Inter-VLAN communication can be controlled using routing policies

Justification:

A hierarchical VLAN-based design provides security, performance, manageability, and scalability. It is therefore suitable for organizations with multiple departments and growing network requirements.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and	Good collaboration	Limited participation	No collaboration or contribution

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		contribution (if team-based)			